

The Universal Digital Notice Board



Session 2024 - 2028

Submitted by:

Muhammad Hassan Raza 2024-CS-213

Supervised by:

Syed Tehseen-ul-Hassan Shah

Course:

CSC-205L COAL LAB

Department of Computer Science

University of Engineering and Technology, Lahore Pakistan

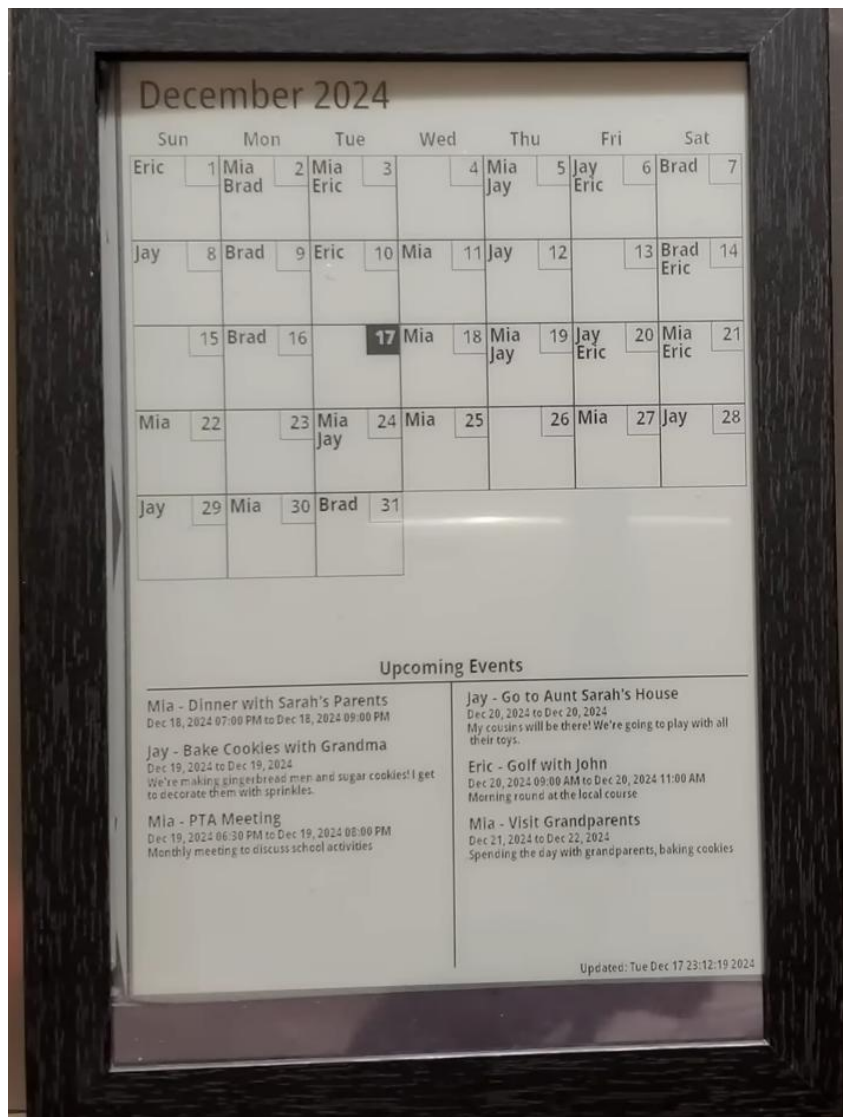
Project Proposal

DeptSync

The Universal Digital Notice Board

1. Executive Summary

This proposal introduces **DeptSync**, a digital replacement for the traditional paper-based notice boards used within the university department. By utilizing an ESP32-controlled e-paper display system, this project aims to transition the department toward a more efficient, environmentally friendly, and real-time communication infrastructure.



2. Problem Statement

The current reliance on physical notice boards presents several operational challenges:

Update Delays

- News and critical tasks are often not posted at the time they are needed, leading to information gaps [User Query].

Resource Inefficiency

- The continuous clipping of paper is not only labor-intensive for staff but also contributes significantly to environmental waste [User Query].

Clutter

- Physical boards become crowded with outdated memos, making it difficult for students and staff to identify the latest updates [User Query].

3. Proposed Solution

The **DeptSync** project proposes a sleek, wall-mounted "Todo Screen" inspired by existing e-paper calendar architectures. This system will serve as a centralized hub for all department news and tasks.

Automatic Synchronization

- The screen will fetch data directly from a digital source (such as Google Calendar or a dedicated app), ensuring that up to six or more upcoming events and tasks are displayed neatly and intuitively.

Effortless Management

- Staff can update information seamlessly through a mobile app or web interface, and the screen will automatically refresh to display the latest content.

Mesmerizing Visuals

- The e-paper screen updates one event at a time, creating a professional and "mesmerizing effect" that draws attention to new updates.

4. Hardware Requirements

To build a robust and energy-efficient system, the following hardware will be required, based on proven ESP32 project designs:

ESP32 Microcontroller

- Acts as the core of the system, handling Wi-Fi connectivity to fetch real-time data from the cloud.

High-Resolution E-Paper Display

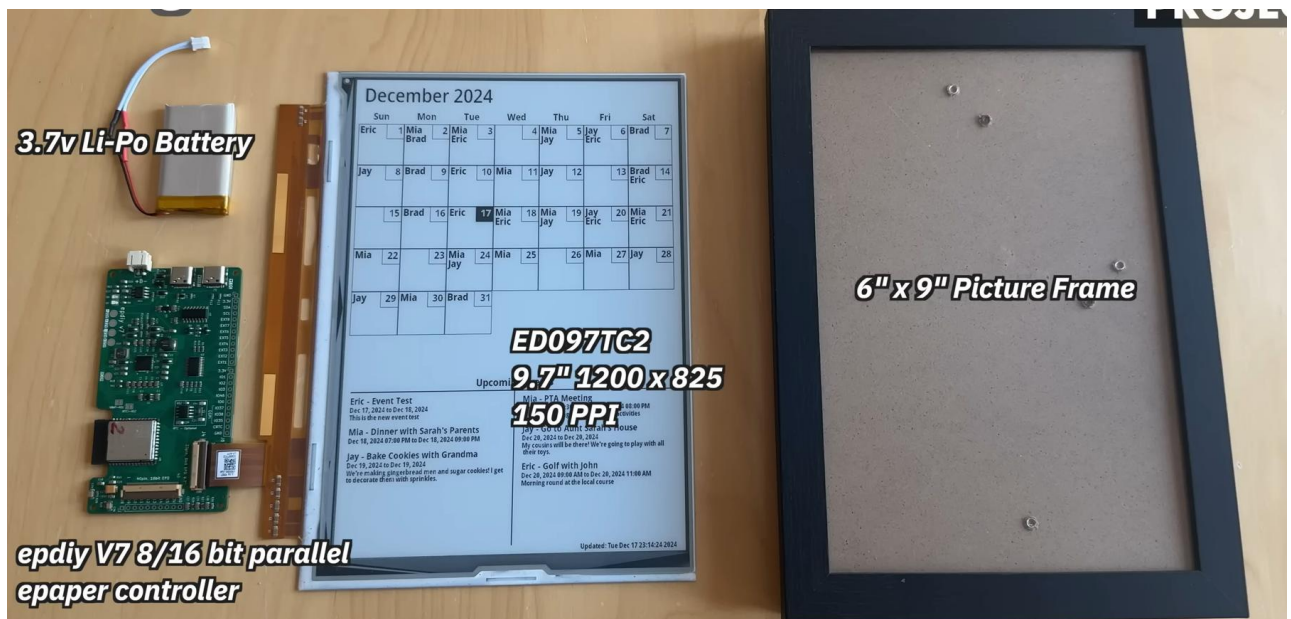
- Selected for its low power consumption and professional, paper-like appearance.

Power Management System

- As e-paper only draws power during refreshes, the device can be battery-powered, allowing for a cordless and clean installation.

Professional Housing

- A repurposed picture frame or custom 3D-printed enclosure will be used to ensure the device matches the department's professional aesthetic.



5. Problem Solved and Benefits

Real-Time Updates

- Eliminates the delay between a news event occurring and its appearance on the board, as the system can be programmed to refresh at specific intervals or immediately upon an update.

Environmental Sustainability

- By removing the need for paper and ink, the department contributes to a "green" campus initiative [User Query].

Operational Efficiency

- Staff are freed from the manual task of clipping papers; a single digital entry from an office computer or phone automatically syncs with the display.

Information Clarity

- The digital layout ensures that only the most relevant, upcoming tasks are visible, reducing cognitive load and information clutter.